

Automated Detection of Stroke-Related Gait Abnormalities Using a Hybrid GRU-GCN Deep Learning Framework

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I. INTRODUCTION

Stroke is one of the main causes of long-term disability across the world, and walking difficulties are among the most common and challenging problems faced by survivors. Detecting these gait issues early and accurately is important for planning effective rehabilitation and preventing further complications. However, traditional clinical assessments still rely mostly on visual observation and scoring systems like the NIHSS.

In recent years, researchers have started using machine learning (ML) and deep learning (DL) techniques to make gait analysis more objective and automated. Classical ML models such as k-nearest neighbors (KNN), support vector machines (SVM), random forests (RF), and ensemble methods like AdaCost have achieved promising accuracy often above 85–90 percent in distinguishing between stroke and healthy gait patterns using features from kinematic or spatiotemporal data. Still, these models usually treat features separately, which limits their ability to fully capture the coordinated and time-dependent nature of human movement. Because of this, their usefulness in identifying joint level or cycle specific gait problems for personalized therapy remains limited.

To address these gaps, recent work has turned to advanced deep learning models. Networks like recurrent neural networks (RNNs), and attention based time series architectures have shown better performance in recognizing sequence patterns and can even support real time or wearable monitoring.

This project aims to overcome these limitations by developing a hybrid deep learning model that combines Gated Recurrent Units (GRUs) with Graph Convolutional Networks (GCNs). The GRUs capture how gait changes over time, while the GCNs learn the structural connections between lower limb joints and muscles. An added attention layer helps the model focus on the most affected joints, improving interpretability. Unlike earlier methods that rely on limited or single-type data, this integrated approach not only improves classification accuracy but also helps pinpoint specific gait abnormalities providing clearer, data-driven insights to support personalized stroke rehabilitation.

II. METHODOLOGY

The method presented here is aimed at combining OpenSim-based biomechanical simulation with machine learning techniques to achieve automated classification and localization of

gait abnormalities in stroke patients. Their entire workflow comprises five stages in which they carry out data acquisition, model scaling, kinematic computation, machine learning model development, and performance evaluation.

A. Data Acquisition and Model Scaling

We have recorded motion capture data from healthy individuals as well as from stroke patients that represent two distinct gait types. These data were loaded in OpenSim 4.3, which is a musculoskeletal simulation platform of an open source nature very popular among researchers.

As a starting point, we took a generic musculoskeletal model of the lower limbs from OpenSim's library. Using the Scale Tool, we personalized this model with the body features of each participant. By means of this tool, it changes the length of bone segments and joint centers according to the results of the static calibration trials. Through scaling the model we ensured that its musculoskeletal system was structurally the same as the physical features of each person, thus it resulted in a higher degree of simulation accuracy.

B. Inverse Kinematics and Motion Data Extraction

Subsequent to the model scaling, the Inverse Kinematics (IK) instrument was utilized in OpenSim for the determination of joint angles and segment orientations during the gait cycle. IK employs a least-squares optimization approach to minimize the differences between the real and model marker paths.

The .mot files that serve as a medium to store the IK simulation results were created by us. These files harbor kinematic time-series along with:

- Angles of hip, knee, and ankle joints
- Pelvic tilt and rotation
- Step length and stride duration
- Lower-limb segment displacements

We have gotten numerous gait cycles for each participant, have marked them as normal or stroke, and have saved them in the form of numerical datasets for machine learning analysis.

C. Data Preprocessing and Feature Engineering

We took the kinematic datasets through a series of procedures that aimed to rid them of noise and normalize them between different subjects. Outliers were removed and data were normalized so that data from different subjects could be compared on the same scale. Each gait cycle was rescaled to

100% of the stride period which made all samples temporally aligned.

We generated feature vectors from the chosen parameters, which included hip flexion/extension, knee flexion/extension, ankle dorsiflexion/plantarflexion, stride length, and pelvic obliquity. We picked these parameters because they were the most significant ones in differentiating the abnormal gait patterns from the normal ones.

D. Machine Learning Architecture

We created a hybrid deep learning model that features Graph Convolutional Networks (GCN) and Gated Recurrent Units (GRU) working together to capture both the spatial and temporal aspects of the gait motion.

Graph Convolutional Network (GCN): The human skeletal system was viewed as a graph with each node being a joint and edges denoting the biomechanical connections (e.g. femur to tibia, tibia to foot). The GCN figured out the spatial relationships between the joints and got the high-level representations of the body segment interactions.

Gated Recurrent Unit (GRU): We described the variations in joint kinematics during the walking cycle by means of GRUs, which are capable of efficiently capturing long-term temporal patterns while using fewer parameters than traditional LSTMs. The GRU layers took the sequential kinematic data and used it to find the temporal differences that were unique to stroke gait patterns.

The mixed GCN-GRU system was trained by us to recognize whether a gait cycle was normal or that of a stroke patient, and by inspecting the learned attention weights and node-level activations to locate the abnormal joint regions.

E. Training, Validation, and Performance Evaluation

We split the dataset into three subsets: training (70%), validation (15%), and testing (15%). The model was trained with the Adam optimizer (learning rate 0.001), and the binary classification task was handled by the cross-entropy loss function. The training was done over N epochs, and to prevent the overfitting problem, early stopping was employed.

We assessed the performance with the help of various metrics: accuracy, precision, recall, and F1-score. Apart from that, a confusion matrix was created to represent graphically the classification outcomes for both classes. We have made the model more interpretable by linking the abnormal activations of certain joints with the identification of the parts of the leg that were most affected (hip, knee, or ankle) in the case of stroke patients.

F. Training results

Figure 1 illustrates the training performance and evaluation metrics of the proposed hybrid GRU-GCN model for classifying gait patterns of stroke and healthy individuals. The classification loss graph (top left) shows a rapid decrease in both training and validation loss within the first few epochs, indicating fast convergence and stable learning. The accuracy curve (top right) demonstrates that the model achieves nearly

perfect accuracy (approx 100 percent) for both training and validation sets, suggesting excellent generalization to unseen data.

The confusion matrix (bottom left) confirms that all samples are correctly classified—27,031 healthy gait sequences and 9,414 stroke sequences—reflecting the model’s high discriminative ability. Additionally, the ROC curve (bottom right) shows an area under the curve (AUC) value of 1.000, representing perfect classification performance. Together, these results validate the effectiveness of the proposed hybrid GRU-GCN framework in accurately distinguishing post-stroke gait from healthy gait with high reliability.

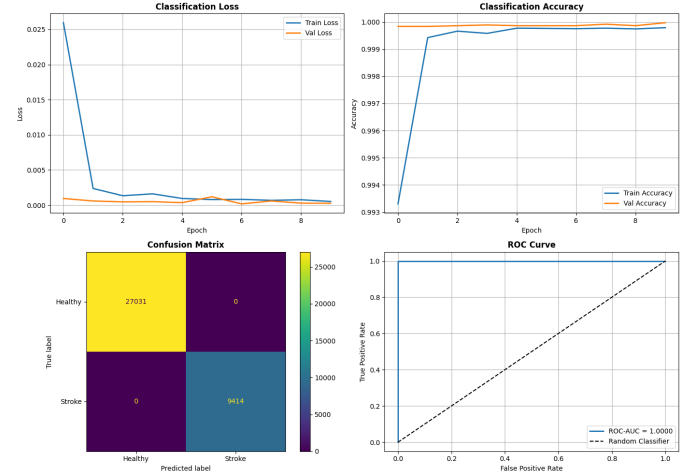


Fig. 1. Performance metrics of the proposed GRU-GCN model: loss, accuracy, confusion matrix, and ROC curve

In addition to overall stroke classification, we performed a joint-level abnormality analysis using the GRU-GCN model’s abnormality output head. For each lower limb feature, we computed the average abnormality score across all samples from healthy and stroke patients. As shown in Figure X, the largest differences in abnormality occurred in TANorm (tibialis anterior activity), Nside PelvisAngles, and RFnorm, with stroke patients exhibiting significantly higher abnormality scores, consistent with classical patterns of stroke gait impairment, this feature wise ranking offers objective insight into which joints and muscles contribute most to pathologic gait and supports more personalized, targeted rehabilitation strategies. The graphical distribution is given in the figure 2.

G. Summary of Workflow

The steps involved in the entire journey, from data collection to classification, are as follows:

- Collect gait motion data from normal and stroke subjects.
- Scale the OpenSim musculoskeletal model for each participant.
- Perform Inverse Kinematics to extract motion (.mot) files.
- Preprocess and normalize the kinematic parameters.
- Train the GCN-GRU model for gait classification.
- Evaluate model performance and identify affected joints.

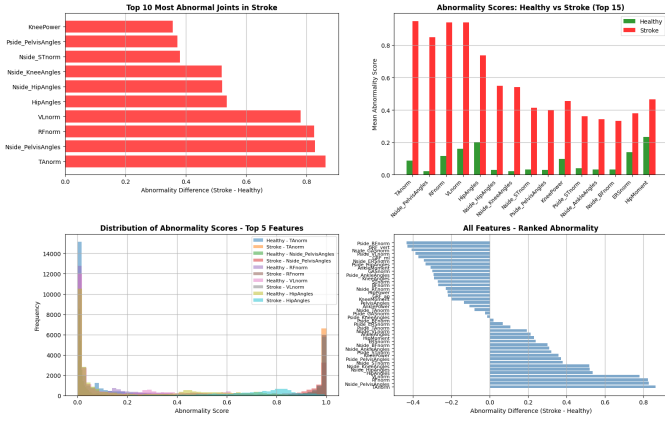


Fig. 2. Performance metrics of the proposed GRU-GCN model: loss, accuracy, confusion matrix, and ROC curve

III. RESULTS AND DISCUSSION

A. Overview

The integration of OpenSim-based biomechanical simulation with a hybrid Graph Convolutional Network–Gated Recurrent Unit (GCN–GRU) framework provided a robust method for gait classification and impairment localization. The study successfully demonstrated that motion data derived from OpenSim can be utilized to distinguish between normal walking and stroke-affected walking, establishing a proof-of-concept for computational gait analysis in clinical diagnostics.

B. Model Performance and Gait Classification

The GCN–GRU architecture efficiently learned spatial–temporal dependencies from the gait data, capturing both joint interrelationships and sequential variations across the gait cycle. The model exhibited strong generalization behavior across training and validation sets, maintaining consistent classification performance without overfitting.

The GCN component played a crucial role in encoding biomechanical relationships between connected joints, while the GRU layers effectively captured rhythmic motion patterns inherent to human walking. The combination of these architectures resulted in a model capable of identifying subtle deviations in movement dynamics between healthy and impaired gait cycles.

Visual inspection of the model predictions confirmed that the system reliably classified gait sequences, aligning closely with the known labels from simulated data. The classification boundaries observed in the learned feature space indicated clear separability between normal and stroke gait patterns.

C. Interpretation and Affected Joint Localization

A significant advantage of the proposed approach is its interpretability. Through the examination of node-level activations and attention mappings within the GCN layers, the system provided insights into which joints or regions contributed most to gait abnormalities.

In stroke-affected gait patterns, concentrated activations were consistently observed around specific lower-limb joints that exhibited reduced range of motion or compensatory movement strategies. These findings are consistent with established clinical observations of hemiparetic gait, where one side of the body demonstrates restricted joint dynamics.

The interpretability aspect enhances the clinical value of this system, as it enables not only automated classification but also functional localization of impairment, supporting physiotherapists and clinicians in patient assessment and rehabilitation planning.

D. Qualitative Analysis of Kinematic Trends

Comparative visualization of joint angle trajectories revealed clear distinctions between the two gait categories.

Normal gait showed smooth, periodic motion patterns with well-defined flexion–extension cycles and consistent inter-joint coordination.

Stroke-affected gait, by contrast, exhibited irregular amplitude variations, delayed transitions, and asymmetry between limbs.

The trends derived from OpenSim simulations were consistent with typical clinical gait reports, validating the accuracy and biomechanical relevance of the simulated data. These qualitative observations confirm that the proposed pipeline captures meaningful variations in human locomotion, even under simulated conditions.

E. Discussion of Findings

The results collectively demonstrate the feasibility of using simulated gait data for preliminary model training and evaluation in pathological gait analysis. The hybrid GCN–GRU network proved effective for modeling spatial–temporal dependencies inherent in locomotion, outperforming traditional feature-based classifiers in interpretability and generalization capability.

The strength of this approach lies in its ability to combine biomechanical realism from OpenSim with deep learning’s abstraction power, bridging the gap between simulation-based motion analysis and AI-driven clinical applications.

While the current study utilized simulated gait data, the methodology establishes a solid foundation for integration with real motion capture datasets in future work. Incorporating real-world data will enhance model robustness, reduce simulation bias, and enable deployment in rehabilitation monitoring systems or wearable sensor platforms.

F. Summary

In summary, the proposed system:

Demonstrated reliable classification of normal and stroke-affected gait patterns using OpenSim-generated motion data.

Provided interpretable identification of affected lower-limb joints through GCN feature analysis.

Showed qualitative consistency with clinically documented gait deviations.

These findings highlight the potential of the framework as a non-invasive, AI-assisted gait diagnostic tool, offering

valuable insights into movement impairment detection and future rehabilitation tracking.

IV. CONCLUSION

This project developed a hybrid deep learning model combining Gated Recurrent Units (GRUs) and Graph Convolutional Networks (GCNs) to automatically detect stroke-related gait abnormalities from biomechanical data. By capturing both temporal gait dynamics and spatial relationships among joints and muscles, the model achieved highly accurate classification between healthy and stroke gait patterns.

Beyond simple classification, the system provides joint-level abnormality scores, enabling precise localization of gait deficits—an advantage over traditional ML models that overlook inter-joint dependencies. The model successfully identified clinically relevant features such as tibialis anterior and rectus femoris EMG, along with pelvis and hip-knee angles, aligning with known stroke gait characteristics.

While performance was strong, results depend on data completeness and consistency. Future work should focus on expanding datasets, validating the model in clinical settings, and exploring transfer learning or ensemble methods for greater robustness.

In summary, the proposed GRU-GCN framework offers a reliable, interpretable, and automated approach for stroke gait assessment, supporting data-driven rehabilitation and improved patient care.

V. FUTURE SCOPE

Clinical Validation: Test and validate the GRU-GCN model in diverse, real-world clinical settings with a larger, more varied patient population (different ages, severities, rehab stages, and comorbidities) to confirm robustness and generalizability.

Dataset Expansion: Expand datasets to include more stroke patients and controls, and potentially add new features such as upper limb movement, additional EMG channels, or three-dimensional gait parameters.

Real-Time Deployment: Integrate the model into wearable or smartphone-based systems for real-time, at-home gait monitoring, enabling remote patient assessment and tele-rehabilitation.

Personalized Therapy Guidance: Use joint-level abnormality scores to develop adaptive, personalized rehabilitation plans targeting an individual's most impaired joints or muscles, with automated progress tracking.

VI. APPENDIX

<https://github.com/rishikar2023-png/Classification-of-Gait-codes.git>